

Feasibility Analysis of Using POF Sensors for Finger Movement in a Soft Hand Exoskeleton

Juan C. Maldonado-Mejía¹, Carlos Arturo Narvaéz Delgado¹, Carlos A. Cifuentes², *Senior Member, IEEE*,
Marcelo Eduardo Vieira Segatto¹, *Member, IEEE*, Camilo A. R. Díaz¹, *Member, IEEE*

Abstract—Measuring finger motion in soft hand exoskeletons remains a challenge due to the nonlinearity and compliance of soft materials. This study evaluates the feasibility of Polymer Optical Fiber (POF) sensors for estimating finger angles in a soft hand exoskeleton. A comparative evaluation was performed against Inertial Measurement Units (IMUs) and a Huskylens camera used as a reference. The test consists of performing repetitions of four movements: depressurized actuator, finger extension, depressurized actuator again, and finger flexion. Two types of analysis were performed: one based on repetitions and another based on the four movements. Results showed that the POF sensor achieved high correlation (r) in both analyses (Repetitions: $r > 0.98$, Movements: $r > 0.85$).

Index Terms—Optical Fiber Sensors, Soft Robotics, Hand Exoskeleton, Inertial Measurement Unit (IMU).

I. INTRODUCTION

The use of robotic devices has significantly transformed hand rehabilitation in post-stroke patients. In particular, hand exoskeletons have increased the training duration and frequency, reducing the labor force and decreasing treatment costs by improving traditional therapy [1]. As a result, these technologies have aided the development of therapies that induce neuroplasticity, which is favourable for patients' recovery [2]. Additionally, hand exoskeletons implement sensors to measure different variables, such as position, angle, velocity, force, torque, and pressure [3]. Thus, the inclusion of sensors facilitates the development of control systems that ensure safe movements and provide information for tracking the patient's progress.

Within robotics technologies, soft hand exoskeletons represent promising devices. Soft robotics integrates compliant and flexible materials, resulting in systems that are lightweight, portable, comfortable, and user-friendly. Thus, hand exoskeletons based on soft robotics have demonstrated their potential to aid in rehabilitation and assist in ADLs, mainly by their ability to perform movements kinematically similar to the natural movements of human joints [4]. Despite their advantages, soft hand exoskeletons are limited in measuring finger joint

movement due to their nonlinear characteristics and infinite degrees of freedom [5].

Some authors have used traditional sensors to measure the joint motion of the fingers. One example is the use of inertial measurement units (IMUs) to acquire trajectories associated with finger movements [6]. Flexion sensors are also employed to predict finger joint stiffness [7] or provide precise tracking and control of the exoskeleton trajectory [8]. However, the stiffness of these may reduce or eliminate the main characteristic of soft exoskeletons. Another approach involves the use of a camera system to capture hand movement, but this kind of system can be sensitive to occlusions or lighting conditions and may require complex setups for precise 3D tracking [9].

Optical fiber sensors, such as Polymer Optical Fiber (POF) sensors, are emerging as a promising solution due to their inherent flexibility, small size, and immunity to electromagnetic interference [10]. POF sensors employ the variation of light intensity to measure stretch, bend, or press variables [11]. For instance, POF sensors have been implemented for angle monitoring in soft prostheses [12], or measurement of the finger angle based on a conical POF configuration [13].

Therefore, this study evaluates the feasibility of using POF sensors for finger motion measurement in a soft hand exoskeleton. This will be achieved by comparing the correlation with other sensors, such as IMUs and a camera. The camera was employed as a reference to assess correlation in motion tracking, rather than as an absolute calibration device.

II. METHODOLOGY

This section describes the experimental setup, including the soft hand exoskeleton, sensor placement, and data acquisition using POF sensors, IMUs, and a camera. Additionally, data processing and analysis were presented.

A. Soft Hand Exoskeleton

The hand exoskeleton is an adaptation of a commercial exoskeleton for stroke rehabilitation (Syrebo, China). The exoskeleton includes a pneumatic system with two air pumps and one solenoid electrovalve. The system was adapted with a network of five solenoid electrovalves (Adafruit Industries LLC, USA), one for each finger, and five air pressure sensors (MPX4250DP, NXP, Netherlands) to measure the air pressure in each actuator. Regarding the electronics, switches based on MOSFET transistors (SI9956DY, Vishay Siliconix, USA) and optocouplers (MCT62S, Onsemi, USA) were implemented to control the pneumatic system. All electronics

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – "CPID 2.2", FAPES/ CNPq/Decit – PPSUS 17/2024), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), and IEEE RAS SPARX.

¹ Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo (UFES), Vitória, Brazil. juan.mejia@edu.ufes.br, carlos.delgado@edu.ufes.br, marcelo.segatto@ufes.br, camilo.diaz@ufes.br

² Bristol Robotics Laboratory, University of the West of England, Bristol, United Kingdom. Carlos.Cifuentes@uwe.ac.uk

were housed within a custom polylactic acid (PLA) 3D-printed control box. The processing and control of the device are performed by a single-board computer (Raspberry Pi 4 Model B) running Ubuntu Server with the Robot Operating System 2 (ROS 2) installed.

B. Sensors

Three sensors are used in this work: POF sensors, IMUs, and a camera. The placement of all sensors was based on the selection of the metacarpophalangeal (MCP) joint, the proximal interphalangeal (PIP) joint, and the fingertip as the primary targets for the analysis of finger movement. This arrangement simplifies the finger kinematics to a two-link, three-point system, in which the finger flexion and extension are represented by the angle formed between these three points. Figure 1 illustrates the placement of these sensors on the exoskeleton.

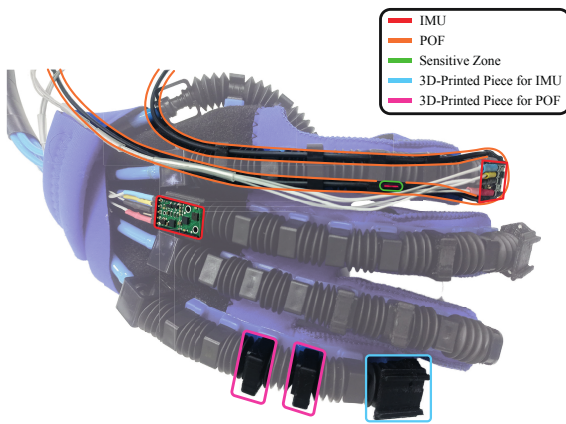


Fig. 1. Sensors placed on the soft hand exoskeleton. The sensors have been highlighted, and the 3D-pieces have been placed on another finger of the exoskeleton to make them more visible.

The following subsections are a detailed description of each sensor, its placement on the exoskeleton, and the data acquisition method.

1) *POF Sensors*: Previous work detailed the manufacture of a POF-based sensor [14]. The optical fiber employed to create the POF sensor was the SH4001 (Mitsubishi Chemical Corporation, USA), a Super Eska Simplex Optical Fiber Cable with a 2.2 mm OD Polyethylene Jacket. This fiber is characterized by a polymethyl-methacrylate resin core with a fluorinated polymer cladding, and a core diameter that ranges from 920 to 1040 μm . To optimize the sensitivity of the POF sensor, a sensitive zone was created by polishing a portion of the fiber core and cladding. The integration of the POF sensor around the index finger of the exoskeleton was made using PLA 3D-printed parts, with the sensitive zone aligned over the PIP joint. The operation of the sensor is based on associating the movement with the variation of the light intensity transmitted through the fiber. Therefore, a system with red light-emitting diodes (IF-E97, Industrial Fiber Optics, USA), phototransistors (IF-D93, Industrial Fiber Optics, USA), and an analog-to-digital converter (ADC) board (ADS1263, Wave-share, China) with 32-bits of resolution was employed to read the values from the POF sensors.

2) *IMUs*: Two IMU sensors (AltIMU-10 v4, Pololu, USA) were used to measure the finger movement. Each IMU integrates a three-axis accelerometer, gyroscope, and magnetometer, providing nine degrees of measurement. These readings are combined to calculate the orientation of each segment as a quaternion representation. The gyroscope provides short-term rotation tracking, while the accelerometer and magnetometer compensate for gyro drift by providing an absolute reference frame. The IMUs were attached to the exoskeleton using a custom PLA 3D-printed piece. An IMU was attached to the fingertip of the index finger of the exoskeleton, and the other was placed on the dorsal side of the exoskeleton. Data communication from the IMUs was performed using the I2C protocol.

3) *Camera*: The Huskylens (DFRobot, China) smart camera was used as a reference. This camera is a module with integrated AI processing and the capability to identify and track objects or tags. For this study, three tags were placed on the exoskeleton: on the fingertip, the PIP region, and the MCP region. The Huskylens provides data for each detected tag, including its ID, X and Y coordinates, and Height (H) and Width (W) in pixels. The camera data was also obtained using the I2C protocol.

C. Experimental Setup and Data Acquisition

The exoskeleton was mounted on a 3D-printed base to stabilize the device during the test. The Huskylens was placed near the exoskeleton to ensure visibility of the three tags placed on the index finger, as shown in Figure 2.

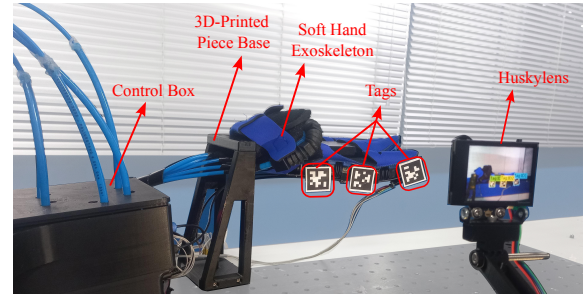


Fig. 2. Experimental setup used for the test.

The test consists of performing nine repetitions of four movements: depressurized actuator, finger extension, depressurized actuator, and finger flexion. The data of the POF sensors, IMUs, and Huskylens were collected simultaneously using ROS 2's *roscat* utility with sampling frequencies of 32.09 Hz, 51.61 Hz, and 6.35 Hz, respectively.

D. Data Processing and Analysis

The collected data from the POF sensors, IMUs, and the Huskylens were processed and analyzed to obtain the index finger movement data and perform the comparison.

1) *Linear Interpolation*: Linear interpolation was used to fill in missing Huskylens data, as the camera occasionally failed to detect one or more tags. Similarly, although data for the IMUs were recorded simultaneously using *roscat*,

timestamp mismatches occurred. Thus, both IMU signals were resampled onto a common time base using linear interpolation.

2) *Relative Depth Estimation Based on Projected Size:* Estimating the relative depth component (Z) is essential for improving the accuracy of angle calculations. The Huskylens provides only two-dimensional coordinates X, Y and bounding box dimensions (H, W) for each tag. Thus, a heuristic method was implemented to estimate Z using H and W as shown in Equation 1.

$$Z = \frac{2WH}{W + H} \quad (1)$$

This approach does not provide an absolute physical distance (e.g., in meters), but preserves the proportional relationship between objects located at different depths in the image.

3) *Angle Calculation and Filtering:* A three-point angle calculation method was applied to obtain the reference angle generated by the Huskylens data. The three points are the tags placed at the MCP, PIP, and fingertip joints defined as P_1 , P_2 , and P_3 , respectively. To determine the joint angle, two vectors are constructed from P_2 to P_1 and another from P_2 to P_3 . Thus, the angle (θ) is obtained using the dot product between these two vectors. To preserve the correct angular orientation, the sign of the cross product is used: if $\vec{P_2P_1} \times \vec{P_2P_3} > 0$, the angle remains unchanged. However, if $\vec{P_2P_1} \times \vec{P_2P_3} < 0$, the final reference angle is $360^\circ - \theta$. In addition, a digital low-pass filter with a cutoff frequency of 1 Hz was applied to the data obtained from the Huskylens and the POF sensors to reduce high-frequency noise. Then, to determine the relative angle between the IMUs, the angular difference was computed using the dot product of the quaternions. The result angle was converted from radians to degrees, and adjusted when necessary to preserve the correct orientation.

4) *Kinematic Analysis and Statistical Metrics:* To evaluate the kinematic performance of the sensors, the signals were interpolated and scaled according to the timestamps of the camera data. Thus, two types of analyses were performed: one based on each repetition and another based on the four movements. In addition, movements are defined as A (depressurized actuator), B (extension), C (depressurized actuator), and D (flexion). The segmentation of these movements was determined using the signal of the pressure sensors of the pneumatic system. The following metrics were used for the evaluation: mean absolute error (MAE), root mean square error (RMSE), bias, Pearson's correlation coefficient (r), mean, standard deviation (STD), and coefficient of variation.

III. RESULTS AND DISCUSSION

Figure 3 shows the processed angular signals from the POF sensors, IMUs, and the camera. In addition, the pressure signals used to segment the four movements were presented.

Pearson's correlation coefficient was calculated to determine the relationship between the angular measurements obtained from the sensors. Figure 4 presents the correlation values. In particular, the camera has the highest correlation pair for both sensors, indicating a correct reference to finger movement.

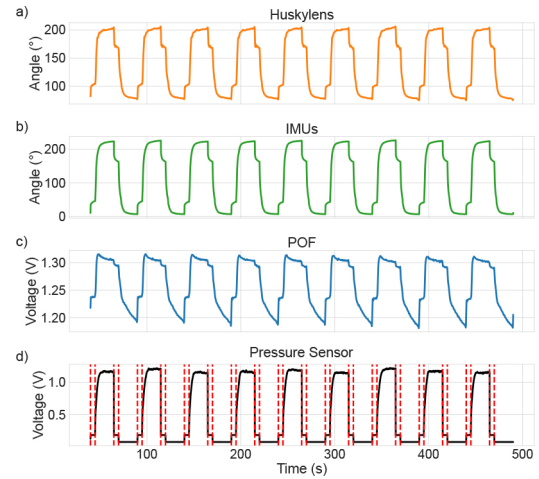


Fig. 3. Processed angular signals obtained from a) Huskylens camera, b) IMUs, and c) the POF sensors. d) The pneumatic pressure signal is also shown, highlighting the motion movements.

Huskylens	1.000	0.999	0.977
IMUs	0.999	1.000	0.973
POF	0.977	0.973	1.000
	Huskylens	IMUs	POF

Fig. 4. Correlation between the three sensors.

Figure 5 shows the signals for repetition 8 after applying interpolation and scaling of POF and IMUs signals regarding the camera signal.

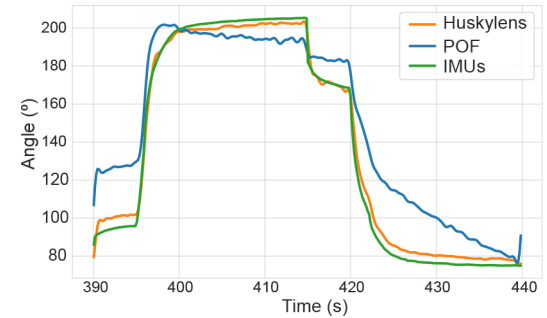


Fig. 5. Time-aligned signals for repetition 8. The signals from the POF sensors and IMUs were interpolated and rescaled according to the camera.

A. Analysis by Repetition

The POF sensor demonstrates adequate motion tracking, as shown in Table I. Nevertheless, it exhibits systematic errors of relatively high magnitude, highlighting the need for proper characterization and calibration. In addition, the correlation of the POF starts to decrease, indicating that a misalignment of the sensor is occurring over the repetitions (see Figure 6). In comparison, IMUs achieve higher accuracy, lower bias, and greater consistency, establishing them as a solid reference.

Even so, the promising performance of the POF sensor indicates that, once properly characterized, it is suitable for use as a sensing technology.

TABLE I
EVALUATION METRICS BY REPETITION FOR POF SENSORS (BLUE ROWS)
AND IMUS (GREEN ROWS).

Metric	Mean $\pm$ STD	CV (%)
MAE ($^{\circ}$)	15.26 $\pm$ 1.21	7.89
RMSE ($^{\circ}$)	18.18 $\pm$ 1.31	7.22
Bias ($^{\circ}$)	12.31 $\pm$ 1.87	15.17
Coefficient r	0.98 $\pm$ 0.00	0.20
MAE ($^{\circ}$)	3.61 $\pm$ 0.06	1.75
RMSE ($^{\circ}$)	4.32 $\pm$ 0.08	1.88
Bias ($^{\circ}$)	-1.70 $\pm$ 0.10	5.72
Coefficient r	1.00 $\pm$ 0.00	0.00

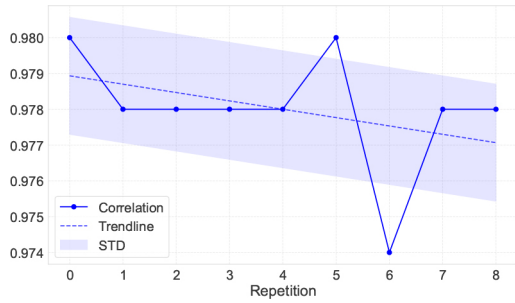


Fig. 6. Correlation coefficients of the POF sensor for each repetition. The IMU correlation data is not presented since the correlation obtained is constant (1.00 ± 0.00).

B. Analysis by Movement

Table II summarizes the performance metrics for each of the movements. For movement B, both sensors showed good performance with low error values. This suggests a more stable or easier-to-measure movement for POF sensors and IMUs. For movements A and D, the POF sensor exhibited a high positive bias, indicating a systematic tendency to overestimate the reference angle. However, the IMUs presented a negative bias, suggesting a slight underestimation in the same segments. Both sensors maintained a high correlation with the Huskylens reference signal across all movements ($r > 0.85$), demonstrating that they adequately captured the temporal trend of finger motion. In particular, the IMUs achieved high correlation values ($r = 1.00$) for movements B and D, indicating excellent tracking of the reference.

TABLE II
EVALUATION METRICS BY REPETITION FOR POF SENSORS (BLUE ROWS)
AND IMUS (GREEN ROWS).

Metric	A	B	C	D
MAE ($^{\circ}$)	27.60 $\pm$ 1.05	8.19 $\pm$ 0.63	13.10 $\pm$ 1.33	19.67 $\pm$ 2.54
RMSE ($^{\circ}$)	27.66 $\pm$ 1.06	10.74 $\pm$ 0.60	13.47 $\pm$ 1.30	21.71 $\pm$ 2.06
Bias ($^{\circ}$)	27.60 $\pm$ 1.05	1.01 $\pm$ 1.68	11.95 $\pm$ 1.67	19.67 $\pm$ 2.55
Coefficient r	0.94 $\pm$ 0.02	0.90 $\pm$ 0.02	0.85 $\pm$ 0.08	0.92 $\pm$ 0.01
MAE ($^{\circ}$)	5.65 $\pm$ 0.14	2.38 $\pm$ 0.10	1.57 $\pm$ 0.28	4.86 $\pm$ 0.14
RMSE ($^{\circ}$)	5.77 $\pm$ 0.13	2.68 $\pm$ 0.10	2.91 $\pm$ 0.79	5.37 $\pm$ 0.17
Bias ($^{\circ}$)	-5.21 $\pm$ 0.15	1.88 $\pm$ 0.12	0.14 $\pm$ 0.57	-4.85 $\pm$ 0.14
Coefficient r	0.93 $\pm$ 0.01	1.00 $\pm$ 0.00	0.93 $\pm$ 0.03	1.00 $\pm$ 0.00

Table III shows that the CV for the MAE of the POF sensor progressively increased from movement A (3.81%) to

D (12.93%). This indicates that the POF sensor is more stable and consistent during movements A and B but becomes less consistent in C and especially in D. In contrast, the IMUs maintained low CV values ($< 5\%$) for both MAE and RMSE in movements A, B, and D, indicating high stability across repetitions. However, for movement C, the CV for MAE increased to 17.60% and for RMSE to 27.19%, highlighting greater variability in the IMU's performance during this movement. This marked difference suggests the presence of outlier errors or spikes in some repetitions of movement C.

TABLE III
CV VALUES OF THE MAE AND RMSE FOR POF SENSORS (BLUE ROWS)
AND IMUS (GREEN ROWS).

Metric	A	B	C	D
CV MAE (%)	3.81	7.74	10.14	12.93
CV RMSE (%)	3.84	5.60	9.69	9.47
CV MAE (%)	2.46	4.09	17.60	2.90
CV RMSE (%)	2.25	3.86	27.19	3.14

Figure 7a illustrates a trend of increasing correlation between the POF sensor and the reference, which could be due to physical settling or progressive stabilization of the sensor itself. Nevertheless, the sensor also presents a high variability between repetitions, which is evidence of its sensitivity to slight structural changes in the system. Conversely, the IMU maintains a more stable correlation throughout the repetitions, although with a slight decrease, possibly attributable to the accumulation of inertial errors.

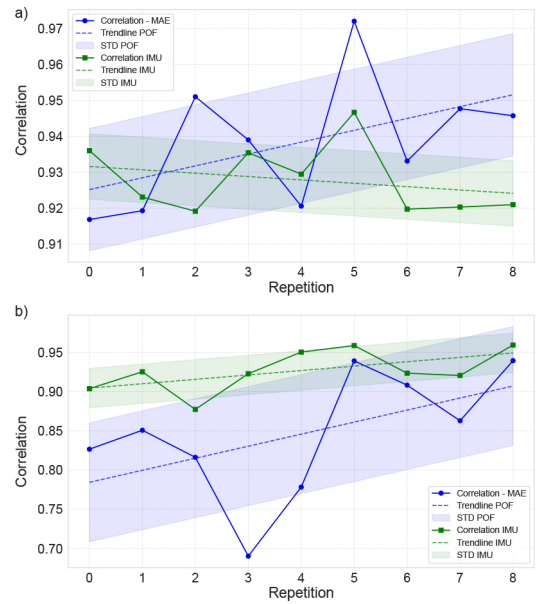


Fig. 7. Correlation coefficients of the POF sensor and IMUs for the a) A movement and b) C movement.

The IMUs are more robust and stable than the POF sensor for C movement, maintaining a high correlation and low variability between repetitions (see Figure 7b). Although the POF sensor also shows a slight improvement with repetitions, its behavior is still more erratic and sensitive to fluctuations. This can be attributed to factors such as internal stresses of the

POF, its viscoelastic material, and nonlinear interaction with the soft exoskeleton.

Furthermore, according to Table III, the IMU shows high coefficients of variation for movement C. However, its strong correlation suggests that much of this apparent variability is due to noise or disturbances from the exoskeleton's actuation that affect the camera reference. In addition, the IMU successfully captures the relative angle. This is evidence of the robustness of the two IMU-based computation, even in the presence of mechanical perturbations of the exoskeleton.

Table IV shows that various sensing technologies have been explored for finger joint movement measurement in robotic gloves. This work shows a contribution in the analysis of the feasibility of POF sensors for finger joint angle estimation in soft hand exoskeletons. The POF sensors achieved a high correlation ($r > 0.85$), highlighting their potential as sensor for these devices.

TABLE IV
SENSORS USED FOR FINGER JOINT MOVEMENT MEASUREMENT.

Reference	Sensor	Main Result
[15]	RFS	Resistive Flex Sensors (RFS)-based gloves showed better reproducibility when the glove was removed and re-worn. The overall average range and SD for RFS-based glove were $5.66^\circ \pm 2.22^\circ$. Overall Intraclass Correlation Coefficient (ICC) for RFS-based glove was 0.73 ± 0.17 .
[16]	FBG	Designed a glove to simultaneously measure the range of motion of ten finger joints with high angular resolution (0.1°). Demonstrated mean error of 0.80° and standard deviation of 1.01° . Fiber Bragg Grating (FBG)-Glove accuracy ($\pm 2^\circ$) is superior to traditional mechanical goniometers ($\pm 5^\circ$).
[15]	IMUs	IMU-based gloves provided better repeatability when the glove was continuously worn. The overall average range and SD for IMU-based glove were $7.80^\circ \pm 2.47^\circ$. Overall ICC for IMU-based glove was 0.76 ± 0.14 .
This work	POF	POF achieved high correlation ($r > 0.85$) with IMUs and the reference camera.

A limitation of this work was the lack of calibration, which may have contributed to systematic mean errors and bias. While the POF sensor achieved high correlation with the reference, proper calibration is essential to enhance accuracy and repeatability. As future work, a calibration procedure based on known bending angles will be implemented to adjust the sensor response and reduce deviations.

Another limitation was the use of the Huskylens camera as a reference. The camera has a low sampling rate and occasionally fails to detect tags. Furthermore, although the inclusion of traditional sensors, such as IMUs, improves the robustness and accuracy of the measurements, their rigid structure can reduce flexibility and softness, intrinsic characteristics of soft exoskeletons. Finally, absolute calibration against a dedicated goniometer or mechanical fixture remains an important step to strengthen the validity of the proposed POF-based sensing approach.

IV. CONCLUSION AND FUTURE WORK

This study demonstrates the feasibility of using POF sensors to measure the movement of the finger in a pneumatic soft hand exoskeleton. The results showed that POF achieved high correlation with both IMUs and the reference camera. Nevertheless, the POF sensor requires further characterization and

calibration, as well as improvements in attachment to reduce misalignment during repeated use. While IMUs remain the most robust option for high-accuracy angular measurements, their rigid structure limits integration in soft devices.

Future work will focus on optimizing sensor integration and calibration, including calibration with a mechanical fixture or a goniometer to complement the correlation-based feasibility demonstrated here. In addition, sensor fusion approaches, such as Kalman filtering, will be considered to combine POF and IMU signals, improving accuracy under noisy conditions. Advanced modeling techniques, such as Nonlinear Autoregressive eXogenous (NARX) models, will also be explored to capture nonlinear relationships in the data. Finally, the system will be validated with users to establish its practical applicability.

REFERENCES

- [1] S. Masiero, P. Poli, G. Rosati, D. Zanutto, M. Iosa, S. Paolucci, and G. Morone, "The value of robotic systems in stroke rehabilitation," *Expert review of medical devices*, 2014.
- [2] M. A. Dimyan and L. G. Cohen, "Neuroplasticity in the context of motor rehabilitation after stroke," *Nature Reviews Neurology*, 2011.
- [3] M. Tiboni, A. Borboni, F. V  rit  , C. Bregoli, and C. Amici, "Sensors and actuation technologies in exoskeletons: A review," *Sensors*, 2022.
- [4] G. M. Whitesides, "Soft robotics," *Angewandte Chemie International Edition*, 2018.
- [5] A. Sarker, T. Ul Islam, and M. R. Islam, "A review on recent trends of bioinspired soft robotics: actuators, control methods, materials selection, sensors, challenges, and future prospects," *Advanced Intelligent Systems*, 2025.
- [6] M. Haghsheenas-Jaryani, C. Pande, and B. M. Wijesundara, "Soft robotic bilateral hand rehabilitation system for fine motor learning," in *2019 IEEE 16th International Conference on Rehabilitation Robotics (ICORR)*. IEEE, 2019.
- [7] K. Li, D. Zhang, Y. Chu, X. Zhao, S. Ren, and X. Hou, "A pneumatic soft exoskeleton system based on segmented composite proprioceptive bending actuators for hand rehabilitation," *Biomimetics*, 2024.
- [8] J. C. Tejada, A. Toro-Ossaba, S. Valencia, N. Gallego, J. J. Jaramillo-Tigres, E. Hernandez-Martinez, and A. L  pez-Gonz  lez, "Soft robotic hand exoskeleton with enhanced pneumatic-type pneumatic actuators for rehabilitation and movement assistance," *Journal of Robotics*, 2024.
- [9] D. Kim, B. B. Kang, K. B. Kim, H. Choi, J. Ha, K.-J. Cho, and S. Jo, "Eyes are faster than hands: A soft wearable robot learns user intention from the egocentric view," *Science Robotics*, 2019.
- [10] L. Bilro, N. Alberto, J. L. Pinto, and R. Nogueira, "Optical sensors based on plastic fibers," *Sensors*, 2012.
- [11] H. Bai, S. Li, J. Barreiros, Y. Tu, C. R. Pollock, and R. F. Shepherd, "Stretchable distributed fiber-optic sensors," *Science*, 2020.
- [12] L. De Arco, M. J. Pontes, M. E. V. Segatto, M. E. Monteiro, C. A. Cifuentes, and C. A. D  az, "Optical fiber angle sensors for the prhand prosthesis: Development and application in grasp types recognition with machine learning," in *2022 IEEE Latin American Electron Devices Conference (LAEDC)*. IEEE, 2022.
- [13] Q.-Q. Ge, T. Zhou, T.-Y. Gong, Y. Liang, L. A. Ngiejunbwen, and M.-Y. Chen, "Highly sensitive measurement of finger joint angle based on a double-u tapered pof embedded in pdms film," *Optical Fiber Technology*, 2023.
- [14] J. C. Maldonado-Mej  a, C. A. Cifuentes, M. E. V. Segatto, and C. A. D  az, "Pof sensor for angle measurement in a textile-based soft hand exoskeleton," in *2024 Latin American Workshop on Optical Fiber Sensors (LAWOFS)*. IEEE, 2024.
- [15] A. Calado, B.-S. Lin, I.-J. Lee, and G. Saggio, "Quasi-static measurement performances of flex-sensor-based and inertial-measurement-unit-based sensory gloves," *IEEE Sensors Journal*, 2023.
- [16] C. K. Jha, K. Gajapure, and A. L. Chakraborty, "Design and evaluation of an fbg sensor-based glove to simultaneously monitor flexure of ten finger joints," *IEEE Sensors Journal*, vol. 21, no. 6, pp. 7620–7630, 2020.